

1. For each of the following subsets of $\mathbb{R}$ find the supremum and the infimum specifying whether these are maximum and minimum respectively.

(a) $A := \left\{ \frac{1+2n}{n} : n \in \mathbb{N} \setminus \{0\} \right\} = \left\{ \frac{1}{n} + 2 : n \in \mathbb{N} \setminus \{0\} \right\};$

The sequence $x_n = \frac{1}{n} + 2$ is decreasing. So,

$$\sup A = 3 = \max A \quad (\text{for } n = 1); \quad \inf A = \lim_{n \rightarrow \infty} \left(\frac{1}{n} + 2 \right) = 2, \text{ no } \min A$$

(b) $B := \left\{ (-1)^n \frac{n+1}{n} : n \in \mathbb{N} \setminus \{0\} \right\};$

All the terms with even indices are positive and all the terms with odd indices are negative. So,

$$\sup B = \sup_{k \in \mathbb{N} \setminus \{0\}} \frac{2k+1}{2k} = \sup_{k \in \mathbb{N} \setminus \{0\}} \left[1 + \frac{1}{2k} \right] = \frac{3}{2} = \max B \quad (\text{for } k = 1 \text{ or } n = 2)$$

$$\inf B = \inf_{k \in \mathbb{N}} \left[-\frac{2k+2}{2k+1} \right] = \inf_{k \in \mathbb{N}} \left[-1 - \frac{1}{2k+1} \right] = -2 = \min B \quad (\text{for } n = 1).$$

(c) $C := \left\{ \frac{m}{n} + \frac{n}{m} : m, n \in \mathbb{N} \setminus \{0\} \right\} = \left\{ \frac{m^2 + n^2}{mn} : m, n \in \mathbb{N} \setminus \{0\} \right\};$

Use $a^2 + b^2 \geq 2ab$ to find that $\frac{m^2 + n^2}{mn} \geq 2$ and hence

$$\inf C \geq 2 \longleftarrow \text{achieved for } m = n = 1. \text{ So,} \\ \inf C = 2 = \min C = 2.$$

On the other hand, $\sup C \geq \frac{1}{n} + n$ for every $n \in \mathbb{N} \setminus \{0\}$ (taking $m = 1$). So, by taking the limit as $n \rightarrow \infty$, we get that $\sup C = \infty$ and hence no $\max C$.

(d) $D := \left\{ \frac{m}{n} - \frac{n}{m} : m, n \in \mathbb{N} \setminus \{0\} \right\};$

Notice that $\inf D \leq \frac{1}{n} - n$ for every $n \in \mathbb{N} \setminus \{0\}$ (taking $m = 1$). So, by taking the limit as $n \rightarrow \infty$, we get that $\inf D = -\infty$ and hence no $\min D$.

On the other hand, $\sup D \geq m - \frac{1}{m}$ for every $m \in \mathbb{N} \setminus \{0\}$ (taking $n = 1$). So, by taking the limit as $m \rightarrow \infty$, we get that $\sup D = \infty$ and hence no $\max D$.

$$(e) \ E := \left\{ \frac{x}{x+1} : x \in (2, 3) \right\} = \left\{ 1 - \frac{1}{x+1} : x \in (2, 3) \right\};$$

The function $f(x) := 1 - \frac{1}{x+1}$ is strictly increasing. So,

$$\sup E = \lim_{x \rightarrow 3^-} f(x) = \frac{3}{4} \quad (\text{no max } E \text{ since } 3 \notin (2, 3))$$

$$\inf E = \lim_{x \rightarrow 2^+} f(x) = \frac{2}{3} \quad (\text{no min } E \text{ since } 2 \notin (2, 3))$$

$$(f) \ F := \left\{ \frac{x}{x-1} : x > 1 \right\} = \left\{ 1 + \frac{1}{x-1} : x > 1 \right\}.$$

The function $g(x) := 1 + \frac{1}{x-1}$ is strictly decreasing. So,

$$\sup E = \lim_{x \rightarrow 1^+} g(x) = \infty \quad (\text{no max } E)$$

$$\inf E = \lim_{x \rightarrow \infty} g(x) = 1 \quad (\text{no min } F \text{ since there is no } x \text{ such that } f(x) = 1)$$

2. Find the following limits:

$$(a) \ \lim_{n \rightarrow \infty} \frac{2}{\sqrt{n + \sqrt{n}} - \sqrt{n - \sqrt{n}}}$$

$$\begin{aligned} \frac{2}{\sqrt{n + \sqrt{n}} - \sqrt{n - \sqrt{n}}} &= \frac{2}{\sqrt{n + \sqrt{n}} - \sqrt{n - \sqrt{n}}} \times \frac{\sqrt{n + \sqrt{n}} + \sqrt{n - \sqrt{n}}}{\sqrt{n + \sqrt{n}} + \sqrt{n - \sqrt{n}}} \\ &= \frac{2(\sqrt{n + \sqrt{n}} + \sqrt{n - \sqrt{n}})}{(\sqrt{n + \sqrt{n}} - \sqrt{n - \sqrt{n}})(\sqrt{n + \sqrt{n}} + \sqrt{n - \sqrt{n}})} \\ &= \frac{2(\sqrt{n + \sqrt{n}} + \sqrt{n - \sqrt{n}})}{n + \sqrt{n} - n + \sqrt{n}} = \frac{2(\sqrt{n + \sqrt{n}} + \sqrt{n - \sqrt{n}})}{2\sqrt{n}} \\ &= \sqrt{1 + \frac{1}{\sqrt{n}}} + \sqrt{1 - \frac{1}{\sqrt{n}}} \end{aligned}$$

Hence,

$$\lim_{n \rightarrow \infty} \frac{2}{\sqrt{n + \sqrt{n}} - \sqrt{n - \sqrt{n}}} = \lim_{n \rightarrow \infty} \sqrt{1 + \frac{1}{\sqrt{n}}} + \sqrt{1 - \frac{1}{\sqrt{n}}} = 2.$$

$$(b) \ \lim_{n \rightarrow \infty} [n \ln(n-1) - (n-1) \ln n].$$

$$\begin{aligned} n \ln(n-1) - (n-1) \ln n &= n \ln(n-1) - n \ln n + \ln n \\ &= \ln(n-1)^n - \ln n^n + \ln n = \ln \left(\frac{(n-1)^n}{n^n} \right) + \ln n \\ &= \ln \left[\left(1 - \frac{1}{n} \right)^n \right] + \ln n \end{aligned}$$

Knowing that $\lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right)^n = e^{-1}$, we get that

$$\lim_{n \rightarrow \infty} [n \ln(n-1) - (n-1) \ln n] = \lim_{n \rightarrow \infty} \ln \left[\left(1 - \frac{1}{n}\right)^n \right] + \lim_{n \rightarrow \infty} \ln n = \infty.$$

3. Prove that if $a_n \rightarrow \ell \in \mathbb{C}$ or $\mathbb{R}$, then $\lim_{n \rightarrow \infty} (a_{n+1} - a_n) = 0$. Is the converse true? Explain your answer.
4. Prove that if $a_n \rightarrow \ell \neq 0$ with $\ell \in \mathbb{C}$ or $\mathbb{R}$, then $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = 1$. Is the converse true? Explain your answer.
5. Show that if $a_1, \dots, a_n$ are n real numbers such that $0 \leq a_i < 1 \forall i$, then

$$\prod_{i=1}^n (1 - x_i) \geq 1 - \sum_{i=1}^n x_i.$$

(Hint: By induction on n !)

6. Show that the sequence (u_n) defined by $u_1 = 0$ and $u_{n+1} = \sqrt{2 + u_n}$ for any $n = 1, 2, \dots$ converges to 2. **Check in your lecture notes!**
7. Study the behaviour (monotonicity, boundedness, convergence) of the sequence of real numbers defined by $a_0 = 1$, and $a_{n+1} = 1 + \frac{1}{a_n}$ for any n .
Hint: Consider the subsequences (a_{2n}) and (a_{2n+1}) .

- (i) First prove by induction that $1 < a_n < 2$ for every $n \in \mathbb{N}$. Hence the sequence is bounded.
- (ii) Now we want to understand why it has been hinted to consider the subsequences (a_{2n}) and (a_{2n+1}) . Knowing that the sequence (a_n) is bounded in order to study its convergence, one would first like to investigate on its monotonicity. Following the examples in the lecture notes we want to study the sign of $a_{n+1} - a_n$ for every n . Notice that

$$a_{n+1} - a_n = \frac{1}{a_n} - \frac{1}{a_{n-1}} = -\frac{a_n - a_{n-1}}{a_n a_{n-1}} \quad (*).$$

Hence, $a_{n+1} - a_n$ and $a_n - a_{n-1}$ have opposite signs since $a_n a_{n-1} > 0$. This seems to suggest that we proceed to the next level in $(*)$ as follows

$$a_{n+1} - a_n = -\frac{a_n - a_{n-1}}{a_n a_{n-1}} = \frac{a_{n-1} - a_{n-2}}{a_n a_{n-1}^2 a_{n-2}} \quad (**)$$

Hence, $a_{n+1} - a_n$ has the same sign as $a_{n-1} - a_{n-2}$ since $a_n a_{n-1}^2 a_{n-2} > 0$.

Therefore, the information regarding the sign of $a_{n+1} - a_n$ is obtained by jumping of two indices and the sign will vary according to whether n is even or odd.

Set $b_n = a_{2n}$ and $c_n = a_{2n+1}$ and let us study the monotonicity of these sequences.

$$\begin{aligned} b_{n+1} - b_n &= a_{2n+2} - a_{2n} = \underbrace{\frac{a_{2n} - a_{2n-2}}{a_{2n+1}a_{2n}^2a_{2n-2}}}_{\text{from (**)}} = \frac{b_n - b_{n-1}}{a_{2n+1}a_{2n}^2a_{2n-2}} = \dots = \frac{b_1 - b_0}{D_n} \\ &= \frac{1}{D_n}(a_2 - a_0) = \frac{1}{D_n a_1} > 0 \end{aligned}$$

where D_n is just a product of positive numbers and hence, is positive. So, $b_{n+1} - b_n > 0$ and therefore, the sequence (b_n) is increasing.

Since

$$c_n = a_{2n+1} = 1 + \frac{1}{a_{2n}} = 1 + \frac{1}{b_n}$$

we get that the sequence (c_n) is decreasing. Since (b_n) and (c_n) are bounded, we have that (b_n) converges to a number ℓ' and (c_n) converges to a number ℓ'' . Now passing to the limit in the equations

$$\begin{cases} c_n = 1 + \frac{1}{b_n} \\ b_n = 1 + \frac{1}{c_{n-1}} \end{cases}$$

we get that

$$\begin{cases} \ell'' = 1 + \frac{1}{\ell'} \\ \ell' = 1 + \frac{1}{\ell''} \end{cases} \Rightarrow \begin{cases} \ell' \ell'' = \ell' + 1 \\ \ell' \ell'' = \ell'' + 1 \end{cases} \Rightarrow \ell' = \ell''.$$

Now, it is your task to prove that the whole sequence (a_n) converges to $\ell' = \ell'' =: \ell$.

Find ℓ !

As you can see the study of a sequence can be tedious!

8. Let (x_n) be a sequence of real numbers defined by $\begin{cases} x_0 \in (0, 1], \\ x_{n+1} = \frac{x_n}{2} + \frac{x_n^2}{4} \end{cases} \quad \forall n \in \mathbb{N}$

- (a) Show that $x_n > 0 \quad \forall n \in \mathbb{N}$. (hint: by induction)
- (b) Show that $x_n \leq 1 \quad \forall n \in \mathbb{N}$. (hint: by induction)
- (c) Show that the sequence (x_n) is monotone (increasing/decreasing).

Notice that

$$x_{n+1} - x_n = \frac{x_n}{4}(x_n - 2) \leq 0 \quad (\text{from (a) and (b)}) \Rightarrow (x_n) \text{ is decreasing.}$$

(d) Deduce that the sequence (x_n) converges and find its limit.

(x_n) converges to some number $\ell \in [0, 1]$ that we find by passing to the limit in the equation $x_{n+1} = \frac{x_n}{2} + \frac{x_n^2}{4}$ and getting $\ell = \frac{\ell}{2} + \frac{\ell^2}{4} \Rightarrow \ell = 0, 2 \Rightarrow \ell = 0$ since $0 \leq \ell \leq 1$.

9. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function. Show that

$$\exists \lim_{x \rightarrow \infty} f(x) = \ell \in \mathbb{R} \Rightarrow \lim_{x \rightarrow \infty} \int_x^{x+1} f(t) dt = \ell.$$

(Hint: You can use the definition of limit at infinity with $\varepsilon > 0$.)

We want to show that

$$\forall \varepsilon > 0 \quad \exists A_\varepsilon > 0 \text{ such that } x > A_\varepsilon \Rightarrow \left| \underbrace{\int_x^{x+1} f(t) dt}_{(*)} - \ell \right| < \varepsilon. \quad (1)$$

The information that we have is that f is continuous and that, for $\varepsilon > 0$,

$$\exists B_\varepsilon > 0 \text{ such that } x > B_\varepsilon \Rightarrow \underbrace{|f(x) - \ell|}_{(**)} < \varepsilon.$$

Now, here is the question: **How do we rewrite/manipulate (*) so that an expression similar to (**) appears?**

Just write $\int_x^{x+1} f(t) dt - \ell = \int_x^{x+1} (f(t) - \ell) dt$ and hence,

$$\left| \int_x^{x+1} f(t) dt - \ell \right| = \left| \int_x^{x+1} (f(t) - \ell) dt \right| \leq \int_x^{x+1} |f(t) - \ell| dt.$$

Now, taking $A_\varepsilon = B_\varepsilon$, we find that

$$\begin{aligned} x > A_\varepsilon &\Rightarrow t > A_\varepsilon \quad \forall t \in [x, x+1] \Rightarrow |f(t) - \ell| < \varepsilon \\ \Rightarrow \left| \int_x^{x+1} f(t) dt - \ell \right| &\leq \int_x^{x+1} |f(t) - \ell| dt \leq \int_x^{x+1} \varepsilon dt = \varepsilon, \end{aligned}$$

and therefore (1) is obtained.

10. Show that $2|xy| \leq \varepsilon x^2 + \frac{1}{\varepsilon} y^2 \quad \forall x, y \in \mathbb{R}, \quad \forall \varepsilon > 0$.

(Hint: Send $2|xy|$ to the right-hand side and identify the identity.)

Here, it is about making the rearrangement so that the inequality becomes $2ab \leq a^2 + b^2$.

11. Show that

$$(i) \quad |\sqrt{|x|} - \sqrt{|y|}| \leq \sqrt{|x - y|} \quad \forall x, y \in \mathbb{R} \quad (ii) \quad |\sqrt{1 + x^2} - \sqrt{1 + y^2}| \leq |x - y| \quad \forall x, y \in \mathbb{R}.$$

(i) was done in the lectures while (ii) is obtained as follows: we first rationalize, that is,

$$\begin{aligned} \sqrt{1 + x^2} - \sqrt{1 + y^2} &= \frac{[\sqrt{1 + x^2} - \sqrt{1 + y^2}] \times [\sqrt{1 + x^2} + \sqrt{1 + y^2}]}{\sqrt{1 + x^2} + \sqrt{1 + y^2}} \\ &= \frac{x^2 - y^2}{\sqrt{1 + x^2} + \sqrt{1 + y^2}} = (x - y) \times \frac{x + y}{\sqrt{1 + x^2} + \sqrt{1 + y^2}} \end{aligned}$$

Hence,

$$\begin{aligned} |\sqrt{1 + x^2} - \sqrt{1 + y^2}| &= |x - y| \times \frac{|x + y|}{\sqrt{1 + x^2} + \sqrt{1 + y^2}} \\ &\leq \underbrace{|x - y| \times \frac{|x| + |y|}{\sqrt{1 + x^2} + \sqrt{1 + y^2}}}_{(\text{using } |x| \leq \sqrt{1 + x^2} \text{ and } |y| \leq \sqrt{1 + y^2})} \leq |x - y| \times \frac{\sqrt{1 + x^2} + \sqrt{1 + y^2}}{\sqrt{1 + x^2} + \sqrt{1 + y^2}} = |x - y|. \end{aligned}$$